

Pathmika Weerarathna

Associate Tech Lead | Full-Stack Engineer (Mobile & Web), AI/ML Exposure

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LinkedIn | GitHub

Summary

Full-stack engineer and Associate Tech Lead with five years of experience delivering production mobile and web applications for telecom operators, having grown through every engineering level at Omobio. Contributed to 20+ client and product engagements across delivery, architecture, and pre-sales in South Asia, Africa, and the Middle East. Core strengths in React Native mobile development (native modules, app security, monetization) and Next.js/React web platforms, backed by REST API and backend engineering. Alongside this, gained hands-on AI/ML exposure self-initiating and leading a telecom fraud-detection R&D initiative spanning NLP embeddings, GenAI, and ensemble ML models. Deepening technical foundations through an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 - Sep 2026

Omobio (Pvt) Ltd

- Led, mentored, and scaled a cross-functional engineering team of 10+ developers, driving solution design and delivery across mobile, web, AI/ML, and data-platform initiatives.
- Served as technical subject-matter expert for pre-sales engagements, producing architecture proposals and technical validation for clients across Africa, South Asia, and the Middle East.
- Led two flagship platforms: the AI/ML R&D team enhancing the SMS Firewall product with ML-driven detection, and the SOR (Steering of Roaming) centralized reporting platform.
- Leveraged AI tools throughout the design and development workflow to boost personal productivity, enabling faster turnaround and higher-quality output.
- Managed client engagements end-to-end and maintained versioned project planning, communications, and technical documentation on company SharePoint for smooth knowledge transfer.

Senior Software Engineer

Jan 2024 - Jan 2026

Omobio (Pvt) Ltd

- Led Agile delivery of end-to-end mobile and web applications, owning architecture design, effort estimation, and SOW reviews.
- Contributed to SQL/CDR reporting engineering for an SMSC billing platform, recovering ~85,000 dropped records and fixing a ~100x query-performance regression via targeted database indexing.
- Developed cross-platform mobile solutions in React Native with native integration (Java, Kotlin, Swift), including a fully custom native alarm engine built from the ground up to replace third-party alarm libraries on Android and iOS.
- Integrated RevenueCat for end-to-end in-app subscription purchasing.
- Self-initiated the AI/ML fraud-detection R&D work now powering the SMS Firewall AI/ML Suite (see Key Projects).
- Led delivery of HOPP, Hutch's Next.js 14 web self-care portal, owning frontend architecture and payment-gateway integration across multiple payment methods, and took it to production.
- Continued ownership of Hutch Selfcare (led for 2+ years across this role and the one prior): led its React Native version upgrade from 0.66.5 to 0.76.5, delivered 3+ change requests for new feature additions, and strengthened its security.
- Mentored a team of 10 junior developers, conducted code reviews, and established coding standards and GitHub Actions CI/CD pipelines.

Software Engineer

May 2022 - Jan 2024

Omobio (Pvt) Ltd

- Led design and development of cross-platform mobile applications (Hutch Selfcare, Vodafone Selfcare, MyDialog, RetailHub, Circadian, SFA) in React Native, managing releases across Play Store, App Store, and Huawei AppGallery.
- Contributed to the React admin-console frontends of Dialog's MDA device-management platform and CDR-disclosure portal, built with Redux state management.
- Hardened Hutch Selfcare against fraud through security optimizations, and led the React Native version upgrade of Vodafone Fiji's production self-care app from RN 0.59.3 to 0.70.5.
- Participated in pre-sales client engagement and contributed to ISO 27001 certification-readiness documentation and a company-wide Git branching standard, later delivered as an internal training program.
- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

- Supported frontend, backend, and mobile feature development for enterprise-grade projects using React.js, React Native, Ext.js/Sencha Touch, and RESTful APIs, including full delivery of a client change request for a production Sales-force mobile application; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- Hutch Selfcare (Telecom Self-Care App), 2+ years:** Owned and evolved Hutch's flagship React Native app: security hardening, dual FCM/HMS push architecture, a RN version upgrade (0.66.5 → 0.76.5), and 3+ feature change requests.
- HOPP Web Self-Care Portal, 2024:** Led the project team; developed a Next.js/TypeScript customer self-care portal with payment-gateway integration, delivered under a Hybrid Git Flow model and deployed to production.
- SMS Firewall AI/ML Suite:** Led a modular AI/ML detection suite (FastAPI, containerized modules, single admin CMS) covering OTP-bypass, phishing-URL, encoded-Unicode, grey-route, and campaign detection; an XGBoost/CatBoost ensemble powers phishing-URL detection, the OTP-bypass/grey-route and encoded-Unicode modules layer in semantic embeddings, and a local-LLM GenAI layer supports content review and translation across the suite.
- SOR (Centralized Reporting Platform):** Led a centralized telecom data-reporting platform spanning backend pipelines and web dashboards, with asynchronous task processing and containerized deployment.
- Circadian (Native Alarm Engine & Monetization), 2024-2025:** Fully custom cross-platform alarm engine (reboot-persistence, background/killed-app handling, astronomy-driven scheduling) with RevenueCat-based subscription and entitlement management.
- SMS Opt-In/Opt-Out Platform (Bangladesh), 2023:** Reviewed the SRS and contributed solution architecture for a language-preference platform: a React/PHP Yii2/Erlang stack serving 5 user roles with low-latency REST APIs.

Additional Experience & Engagements

Separately architected FirewallBI, the telecom BI platform underpinning the AI/ML suite above (2023-2025; centralized on-prem deployment, hundreds of Logstash pipelines, mutual-TLS Elasticsearch cluster), and contributed to further engagements, including Dialog's CMDP, RetailHub, WinBack, and DF platforms and the RoamerSteering ELK analytics platform (2023), plus pre-sales solution proposals for operators across South Asia and Africa.

Areas of Expertise

- Mobile:** React Native (incl. New Architecture), Android (Java/Kotlin), iOS (Swift/Objective-C), Huawei HMS, native module bridging
- Web:** React.js, Next.js, TypeScript/JavaScript, Redux
- Backend & APIs:** Python, FastAPI, REST API design, Microservices, Java (Spring Boot), .NET (ASP.NET Core)
- AI/ML (exposure):** PyTorch, Hugging Face Transformers, LangChain, XGBoost, CatBoost, SHAP, NLP, Semantic Search
- Data Engineering:** PostgreSQL, ClickHouse, MariaDB/MySQL, Elasticsearch/ELK, Apache Airflow, Celery
- DevOps:** Microsoft Azure (AZ-900 Certified), Docker, Git, CI/CD (GitHub Actions, GitLab CI, Jenkins), Linux, Prometheus, Grafana
- Leadership:** Agile delivery, technical mentoring, pre-sales & client engagement, architecture ownership

Education

Master of Science in Artificial Intelligence (Reading) - University of Moratuwa July 2025 - Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 - University of Colombo School of Computing 2019 - 2022

Diploma in English (Certificate Level), Distinction Pass - Seagis Campus, Nugegoda

2018

Certifications

Microsoft Certified: Azure Fundamentals (AZ-900) - [Verify Credential](#)

Aug 2026

Achievements

- Emerging Employee of the Year - Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee - University of Colombo (2021): achieved GPA above 3.7 in first academic year.